Evaluation of PQC-based System Availability in Iridium Satellite Communications*

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Abstract

Given the threat posed by quantum computing and the long operational lifespan of satellite systems, examining the applicability of Post-Quantum Cryptography (PQC) over low-data-rate satellite links is an urgent challenge. This paper deploys NIST-standardized PQC algorithms on an actual Iridium SBD link and empirically analyzes the latency incurred under strict payload constraints.

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1 Introduction

The threat of quantum computing to classical public-key cryptography has led NIST to establish ML-KEM(FIPS 203) and ML-DSA(FIPS 204) as quantum-resistant replacement standards[1, 2]. While computational performance on embedded hardware has been sufficiently studied, deployment over low-data-rate satellite remains largely unexplored. Satellite systems currently in operation have a design life of 10 to 20 years and are directly exposed to "Harvest Now, Decrypt Later, HNDL" threats.

Iridium, which operates 66 LEO satellites and is used for military/defense command-and-control and IoT/industrial device services, imposes payload limitations on transmission data [3].

The Iridium satellite communication environment has many restrictions to immediately consider PQC migration. Therefore, this paper provides an empirical delay time profile of ML-KEM-based key exchange and ML-DSA-based digital signature system on an Iridium SBD in operation.

2 PQC Application Model in Satellite Communications

Figure 1 illustrates a system that performs data integrity verification and key exchange using ML-DSA and ML- KEM. The figure depicts the overall data flow structure; due to the payload constraints of Iridium satellite communication, fragmentation of data is unavoidable.

In this experiment, the Iridium modem is an AT command-controlled 9602 SBD that can actually transmit up to 340 bytes(transmission) and 270 bytes(reception) packets, but it was conducted in an environment limited to 200 ASCII characters [3]. The PQC data below is converted to hexadecimal to double the number of bytes, and the effective binary payload per piece is reduced to 100 bytes.

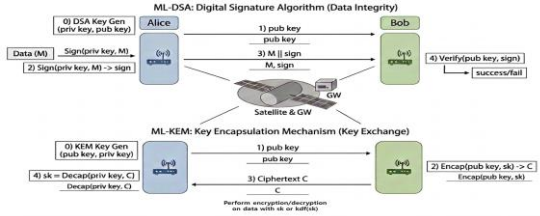


Figure 1: PQC-based system in satellite environment

ML-KEM key exchange performs 1) transmit public key over [Public key length/100], (2) ciphertext encapsulated with public key, 3) Transmission of ciphertext over [ciphertext length/100], and 4) Decapsulate with private key to calculate shared key. ML-DSA data integrity verification performs 1) public key transmission over [public key length/100], 2) data signature generation, 3) data over [data + signature length/100], and 4) signature verification.

3 Experimental Results

The results for ML-KEM-512 and ML-DSA-44, which correspond to security level 1 for ML-KEM and ML-DSA, are as follows. For ML-KEM-512, the cumulative RTT of public key transfer (8 fragments) is 98.337 seconds, the RTT of ciphertext transfer (8 fragments) is 111.041 seconds, and the initial key exchange delay time is 209.4 seconds. The RTT of ML-DSA-44 public key (14 fragments) takes 189.98 seconds, data|||signatures (25 fragments) take 339.29 seconds. As above, the long delay time is mainly due to fragmentation for small iridium SBD payloads.

4 Conclusion

This paper identified the first empirical delay time for ML-KEM and ML-DSA on an iridium SBD link that is actually in operation. The measured key exchange and signature delay time is considerable and is not suitable for interactive use, but it is not advisable to abandon the introduction of PQC based on this. The HNDL threat is a realistic risk considering the operational lifespan of the satellite system. Since public key distribution and session key sharing occur once per session, this cost is sufficiently tolerable in a store-and-forward IoT operation environment. Future research will pursue lightweight encoding, fragmentation reduction, and PQC hybrid methods to further reduce satellite PQC overhead.

References

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